FIFA 21 DATA CLEANING AND ANALYSIS

Data Quality, Player Attributes, Position Analysis, and Regression Modeling

Prepared using R | Dataset: FIFA 21 player data

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1. Executive Summary

This project uses R to clean and analyze the FIFA 21 player dataset. The workflow includes standardizing text fields, converting height and weight to numeric measurements, parsing financial fields, converting selected rating variables, processing club-joining dates, checking data quality, summarizing player attributes by position, identifying high-market-value players with relatively low wages, and fitting simple and multiple linear regression models.

Two regression findings are currently available. In the simple regression of Interceptions on Height, the height coefficient is -0.461 ($p = 1.74 \times 10^{-16}$), while $R^2 = 0.00357$. This indicates a statistically significant negative association, but height alone explains very little variation in interceptions. In the multiple regression for Agility, the six predictors explain approximately 77.0% of the observed variation ($R^2 = 0.770$; adjusted $R^2 = 0.770$; n = 18,979). These are associations, not evidence of causation.

2. Project Objectives and Questions

- Improve the usability and consistency of the raw FIFA 21 dataset.
- Check data structure, missing values, and residual formatting issues.
- Compare average height, weight, and selected player attributes across positions.
- Identify players in the upper quartile of market value and lower quartile of wage.
- Investigate the relationship between height and interceptions.
- Investigate the relationship between weight and agility.
- Evaluate a multiple regression model using six player attributes to explain agility.

3. Dataset and Tools

The analysis starts by reading the raw CSV file into fifa_raw and creating a working copy named data. The workflow uses R and packages from the tidyverse ecosystem, including dplyr, readr, lubridate, broom, scales, janitor, and ggrepel. The regression tables are assembled with broom, visualizations are created with ggplot2, and result tables can be exported to Excel using writexl.

	Metric	Description
1	Number of rows	18979
2	Number of columns	81
3	Key variables	Name, Age, Height, Weight, Value, Wage, BP, Dribbling, Agility, Interceptions

Figure 1. Overview of the FIFA 21 Dataset

4. Data Cleaning

The raw data were copied before transformation so the original imported dataset remained available for comparison. The cleaning workflow applied the following transformations:

Cleaning step	Method
Text normalization	Replaced line breaks, carriage returns, and tabs in character columns with spaces, then squished repeated whitespace.
Height	Removed the inch symbol and converted the feet-and-inches text representation to total inches.
Weight	Removed the 'lbs' suffix, converted the value to numeric, and converted pounds to kilograms using 2.20462 pounds per kilogram.
Financial fields	Removed currency symbols and converted K/M suffixes into numeric multipliers for Value, Wage, and Release.Clause.
Star/rating fields	Removed star symbols and whitespace from W.F, SM, and IR, then converted the fields to numeric.
Joined date	Parsed Joined into a date using the specified date formats.
Years at club	Calculated years between Joined and the fixed reference date 2020-10-01.

5. Data Quality Checks

The code inspects the dataset structure with glimpse(data), counts missing values by column with colSums(is.na(data)), and checks whether newline/tab characters or unnecessary whitespace remain in character columns. These checks help identify potential data-quality issues after the initial cleaning.

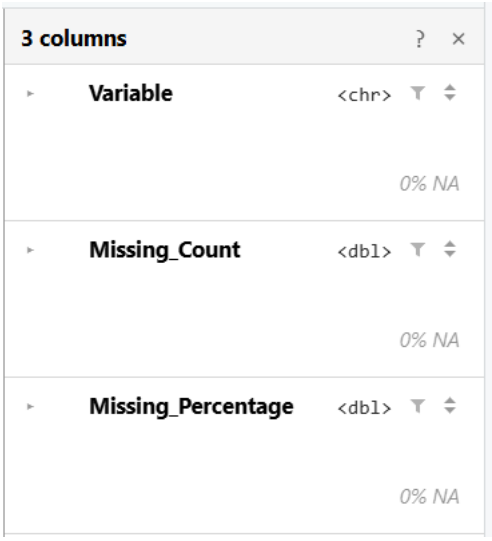


Figure 2. Missing Values Summary

Note: This screenshot captured only the column headers of the missing-value summary table; the data rows (Missing_Count and Missing_Percentage for each variable) are not visible. This figure should be recaptured from the RStudio Viewer showing the populated table before final submission.

	Newline/Tab Issues	Extra Whitespace Issues
1	0	0

Figure 3. Results of Residual Whitespace Checks

6. Descriptive Analysis by Position

6.1 Average height and weight

The analysis calculates average height in inches and average weight in kilograms by the primary position variable BP, excluding goalkeepers and missing positions for the initial descriptive tables. The resulting tables are ordered from highest to lowest average values.

	BP	average_height	number_of_players
1	CB	73.23582	3685
2	ST	71.98843	2680
3	CDM	70.93564	1445
4	CM	70.56160	1047
5	LB	70.25046	1086
6	RB	70.14537	1080
7	LWB	69.96169	261
8	RWB	69.88448	277
9	CF	69.70513	78
10	CAM	69.60026	2299
11	RM	69.46058	1611
12	LM	69.45006	871
13	LW	69.29570	186
14	RW	69.26174	298

Figure 4. Average Height by Position

	BP	average_weight	number_of_players
1	CB	79.27930	3685
2	ST	77.09810	2680
3	CDM	74.38641	1445
4	CF	73.07498	78
5	CM	73.01243	1047
6	LB	72.44873	1086
7	RB	72.20863	1080
8	RWB	71.80032	277
9	LWB	71.41395	261
10	LM	70.67561	871
11	LW	70.56540	186
12	RW	70.46216	298
13	CAM	70.33768	2299
14	RM	70.27903	1611

Figure 5. Average Weight by Position

6.2 Players with more than ten years at their club

The long_term_players table filters players whose calculated years_at_club exceeds ten years and displays their names, primary positions, joining dates, and calculated tenure. Because tenure is measured relative to October 1, 2020, this interpretation depends on that reference date.

	Name	BP	Joined	years_at_club
1	H. Sogahata	GK	1998-01-01	22.74863
2	I. Akinfeev	GK	2002-01-01	18.74863
3	Kim Kwang Suk	CB	2002-01-01	18.74863
4	D. Lewington	LB	2002-08-01	18.16712
5	K. Nakamura	CM	2003-01-01	17.74863
6	H. Otani	CDM	2003-01-01	17.74863
7	M. McNulty	GK	2003-01-01	17.74863
8	P. Karlsson	CB	2003-07-01	17.25205
9	D. Zibung	GK	2003-07-01	17.25205
10	M. Bloomfield	CM	2003-12-22	16.77596
11	T. Aoyama	CM	2004-01-01	16.74863
12	Yang Qipeng	GK	2004-01-01	16.74863
13	L. Messi	RW	2004-07-01	16.25205
14	I. Khune	GK	2004-07-01	16.25205
15	A. Considine	CB	2004-07-01	16.25205
16	P. Netzer	CB	2004-07-01	16.25205
17	M. Noble	CM	2004-08-01	16.16712
18	A. Aguilmera	CB	2005-01-01	15.74863

Showing 1 to 17 of 204 entries, 4 total columns

Figure 6. Players with More Than 10 Years at Their Clubs

7. High-Value, Low-Wage Player Screening

The analysis defines a high-value, low-wage group using distribution-based thresholds: market value at or above the 75th percentile and wage at or below the 25th percentile. It then calculates Value_Wage_Gap as market value minus wage and Wage_to_Value as wage divided by value. The ten highest-value players within this selected group are displayed and visualized.

This is a screening rule for further investigation, not proof that a player is underpaid. Market value and wage have different meanings and time scales, so their difference should not be interpreted as a literal financial loss or compensation gap without additional context.

	Name	Positions	Value	Wage	Value_Wage_Gap	Wage_to_Value
1	Pau López	GK	25500000	500	25499500	1.960784e-05
2	V. Tsygankov	RM	20500000	500	20499500	2.439024e-05
3	L. Pellegrini	CAM CM CDM	17000000	500	16999500	2.941176e-05
4	D. Livaković	GK	14500000	500	14499500	3.448276e-05
5	Ismaily	LB	14000000	500	13999500	3.571429e-05
6	Taison	LM CAM	14000000	500	13999500	3.571429e-05
7	Marlos	RM CAM	14000000	500	13999500	3.571429e-05
8	E. Džeko	ST	13500000	500	13499500	3.703704e-05
9	J. Veretout	CDM CM	13500000	500	13499500	3.703704e-05
10	H. Mkhitaryan	RM CAM RW	13000000	500	12999500	3.846154e-05

Figure 7. Top 10 Players with High Market Value and Low Wages

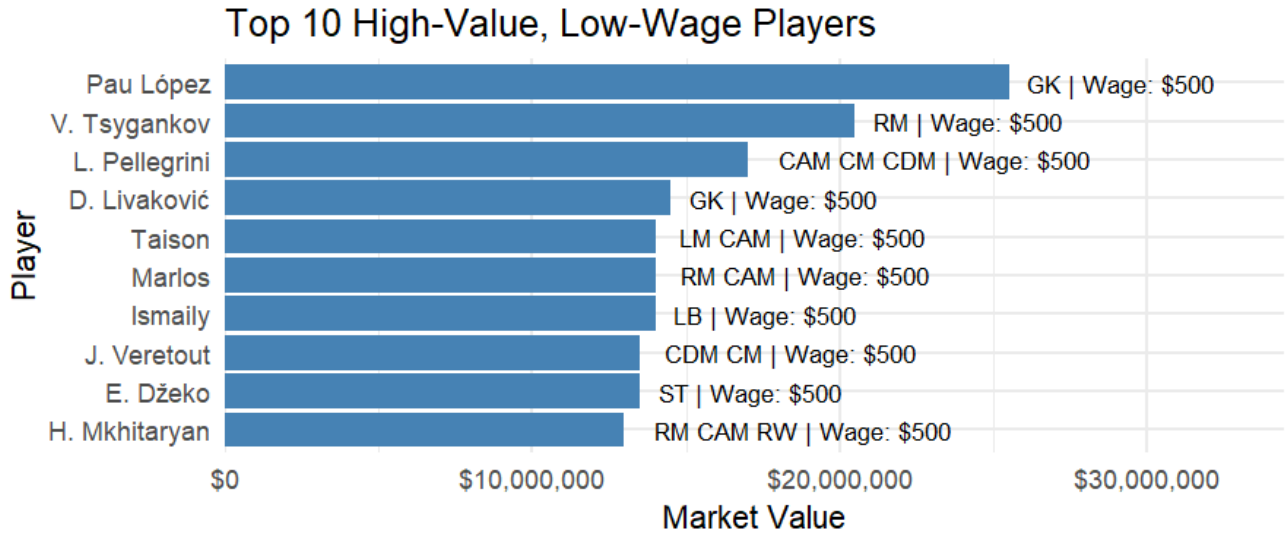


Figure 8. Top 10 High-Value, Low-Wage Players (Bar Chart)

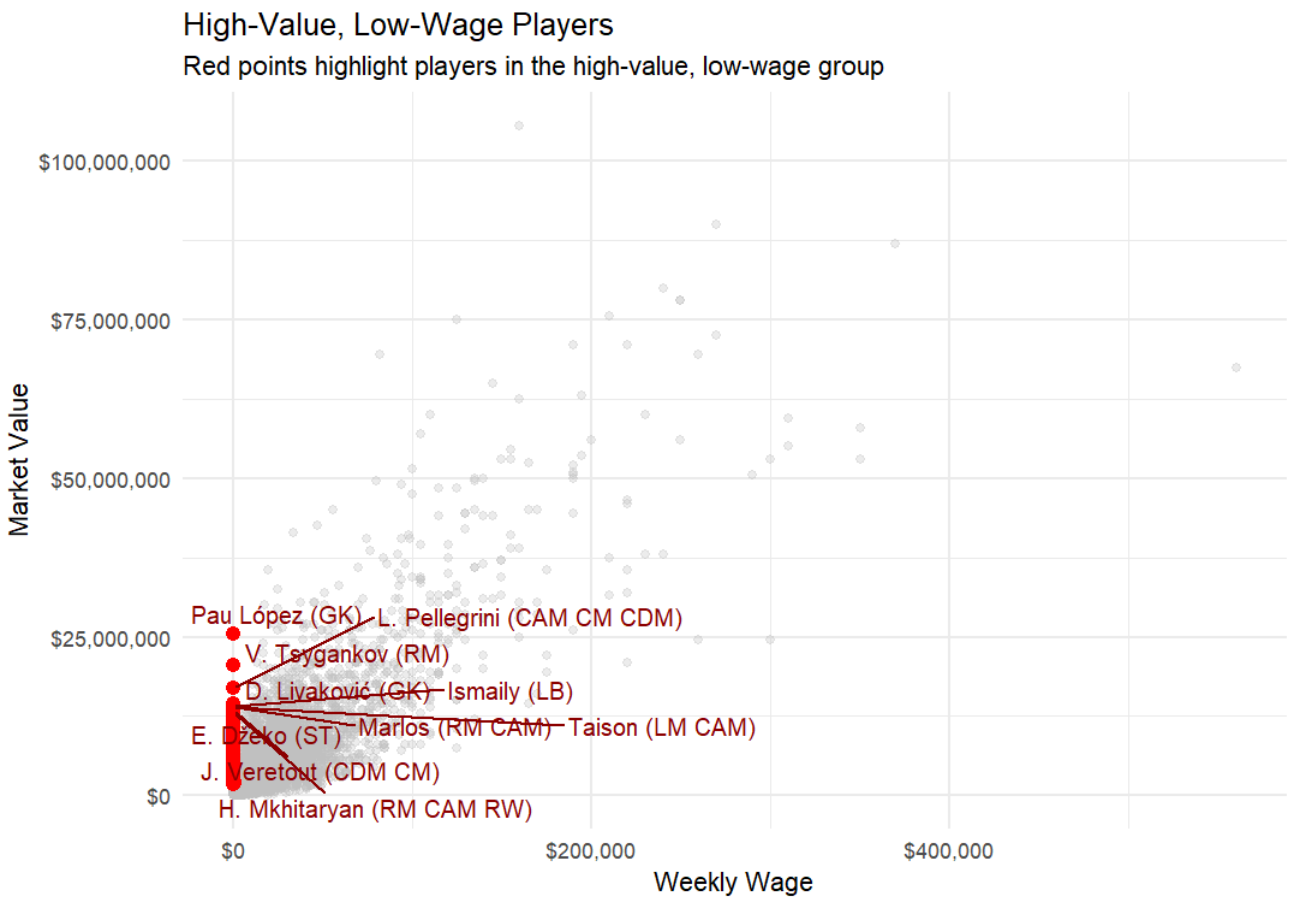


Figure 9. Market Value versus Weekly Wage, Highlighting the High-Value, Low-Wage Player Group (Scatter Plot)

8. Dummy Variables and Position Analysis

Three binary indicators are created: Left identifies left-footed players, Striker identifies players whose primary position is ST, and Goalie identifies players whose primary position is GK. A model matrix is also created to generate indicator columns for the BP position categories. The subsequent field-player analysis excludes goalkeepers.

The combined position_summary table contains player counts and average height, weight, dribbling, attacking, agility, and wage by position. The comparison_table identifies the positions with the highest and lowest average values for dribbling, attacking, agility, and wage. Note that the Average_Attacking values (roughly 230–327) are well above the usual 0–100 scale used for individual FIFA attributes such as Agility and Dribbling, which suggests Average_Attacking may be a summed composite of several attacking sub-attributes rather than a single rating; this should be confirmed before comparing it directly to Agility and Dribbling in Figure 12.

	BP	Players	Average_Height	Average_Weight	Average_Dribbling	Average_Attacking	Average_Agility	Average_Wage
1	CAM	2299	69.60026	70.33768	67.80296	284.7869	72.95302	9024.598
2	CB	3685	73.23582	79.27930	46.39620	230.0019	53.83935	8184.708
3	CDM	1445	70.93564	74.38641	61.01384	269.9931	64.56817	9562.526
4	CF	78	69.70513	73.07498	73.29487	327.4744	74.79487	23444.872
5	CM	1047	70.56160	73.01243	65.76982	291.1089	67.28653	12953.820
6	LB	1086	70.25046	72.44873	61.43278	258.5599	68.18232	8552.118
7	LM	871	69.45006	70.67561	67.99885	287.8691	74.84501	8574.971
8	LW	186	69.29570	70.56540	71.74194	300.5484	78.81720	16202.151
9	LWB	261	69.96169	71.41395	64.10345	263.0651	71.65134	8256.130
10	RB	1080	70.14537	72.20863	61.22685	258.4065	68.96111	8864.074
11	RM	1611	69.46058	70.27903	66.50217	279.1092	74.22595	6832.309
12	RW	298	69.26174	70.46216	71.12752	301.5000	78.95973	14585.403
13	RWB	277	69.88448	71.80032	64.56679	263.1119	71.43682	7955.415
14	ST	2680	71.98843	77.09810	63.14776	295.9030	65.34664	9144.925

Figure 10. Average Height, Weight, Dribbling, Attacking, Agility, and Wage by Position (position_summary Table)

	Metric	Best_Position	Best_Average	Worst_Position	Worst_Average
1	Dribbling	CF	73.29487	CB	46.39620
2	Attacking	CF	327.47436	CB	230.00190
3	Agility	RW	78.95973	CB	53.83935
4	Wage	CF	23444.87179	RM	6832.30912

Figure 11. comparison_table

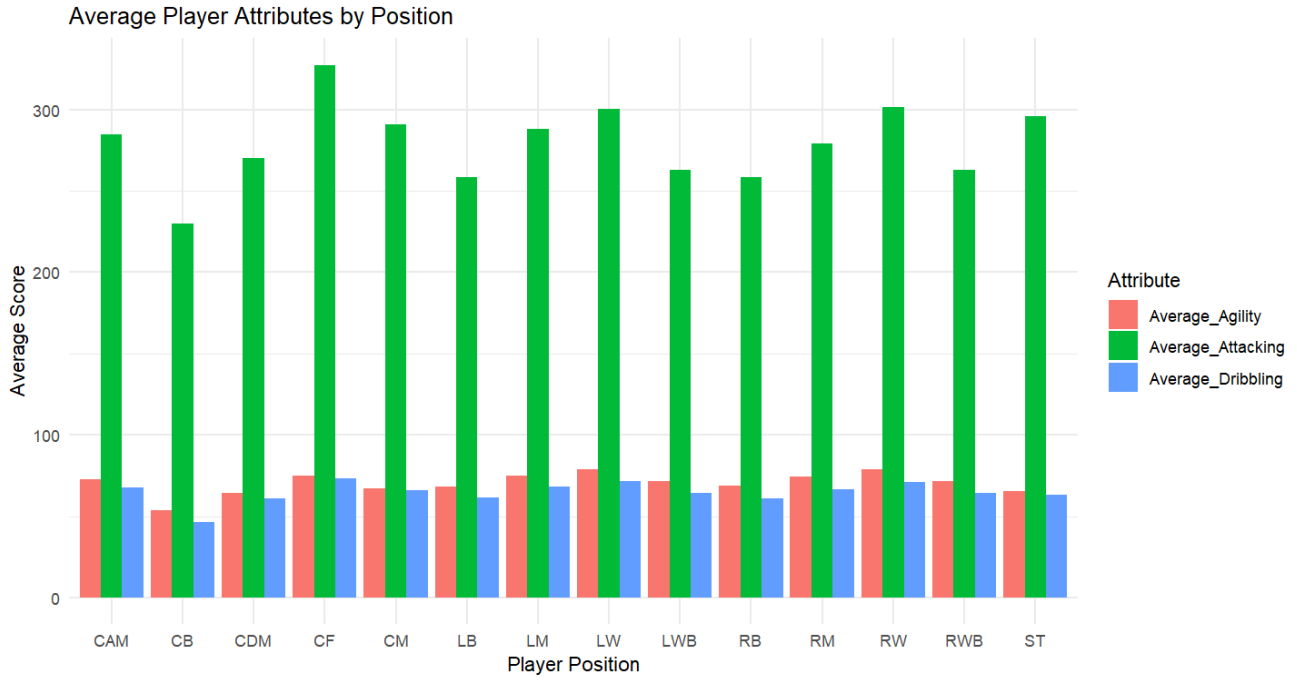


Figure 12. Comparison of Average Dribbling, Attacking, and Agility by Position (Grouped Bar Chart)

9. Regression Analysis

9.1 Height and Interceptions

A simple linear regression was fitted with Interceptions as the outcome and Height (inches) as the predictor. The estimated coefficient for Height was -0.461, with standard error 0.0559, t = -8.25, and p = 1.74×10^{-16}. The model R^2 was 0.00357, adjusted R^2 was 0.00352, and the sample size was 18,979.

Interpretation: each additional inch of height was associated with an estimated 0.461-point decrease in interceptions on average in this simple linear model. The association was statistically significant, but the low R^2 shows that height alone explains only about 0.357% of the observed variation in interceptions.

	Relationship	term	estimate	std.error	statistic	p.value	R_squared	Adjusted_R_squared	Model_p_value	Observations
1	Height -> Interceptions	(Intercept)	79.2617426	3.9886589	19.871778	5.474091e-87	0.003570407	0.003517899	1.741903e-16	18979
2	Height -> Interceptions	Height	-0.4606666	0.0558647	-8.246113	1.741903e-16	0.003570407	0.003517899	1.741903e-16	18979

Figure 13. Relationship Between Player Height and Interceptions

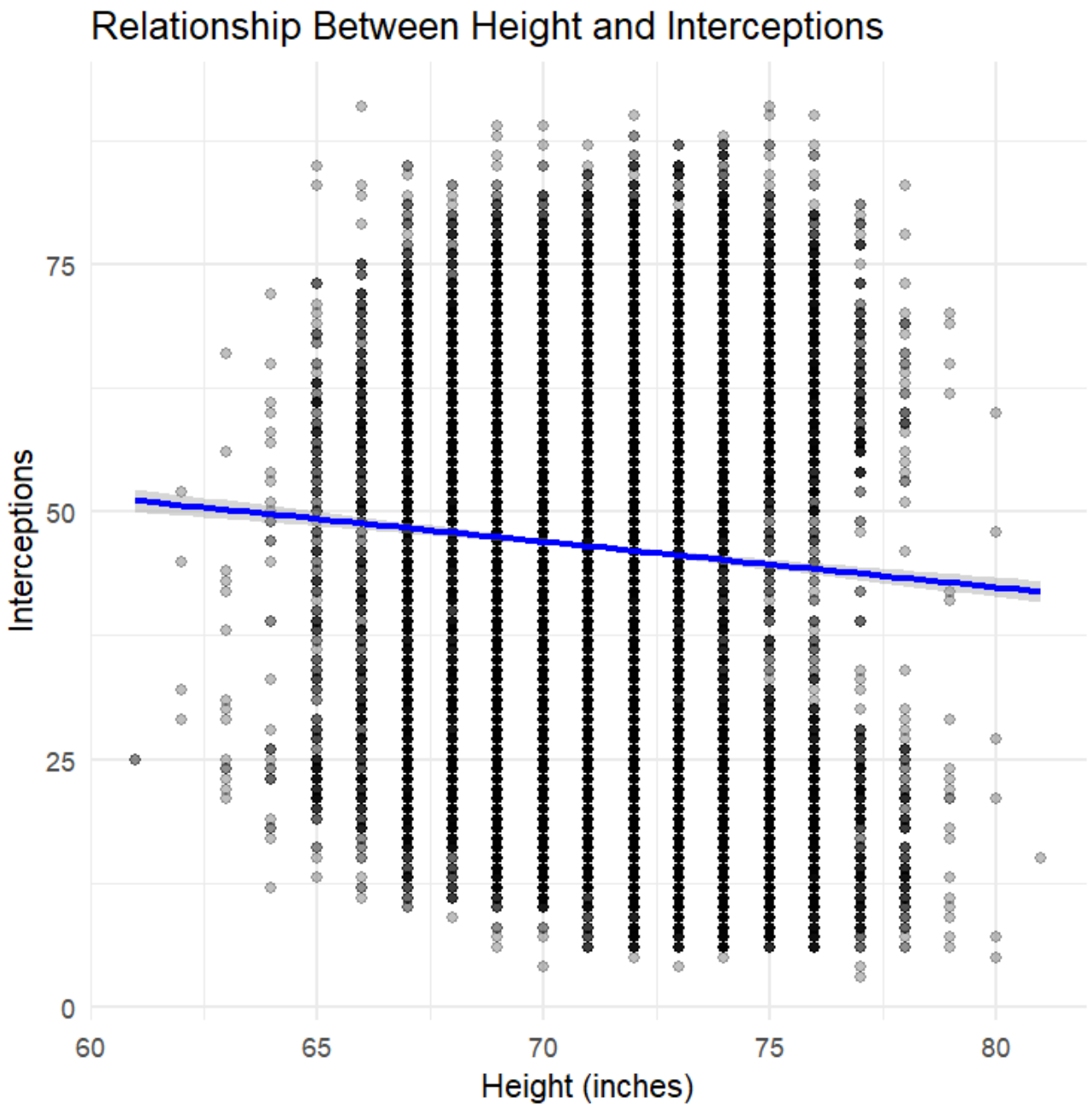


Figure 14. Scatter Plot of Height versus Interceptions with Fitted Regression Line
The estimated coefficient for Height was negative (-0.4607), indicating a negative association between player height and interceptions. The p-value was below 0.05 ($p < 0.001$), indicating that the relationship was statistically significant. The model had an R^2 of 0.00357, meaning that Height explained approximately 0.357% of the variation in Interceptions.

9.2 Weight and Agility

A simple linear regression was fitted with Agility as the outcome and Weight (kilograms) as the predictor. The resulting coefficient table and scatter plot are shown below, followed by an interpretation of the estimated relationship.

	Relationship	term	estimate	std.error	statistic	p.value	R_squared	Adjusted_R_squared	Model_p_value	Observations
1	Weight -> Agility	(Intercept)	145.513410	0.95731890	152.00098	0	0.2813226	0.2812847	0	18979
2	Weight -> Agility	Weight	-1.095003	0.01270476	-86.18844	0	0.2813226	0.2812847	0	18979

Figure 15. Linear Regression Results: Weight and Agility



Figure 16. Scatter Plot of Weight versus Agility with Fitted Regression Line

The estimated coefficient for Weight was negative (-1.095), indicating a negative association between weight and agility. The p-value was below 0.05, indicating that the relationship was statistically significant. The model had an R^2 of 0.2813, meaning that Weight explained approximately 28.13% of the variation in Agility.

9.3 Multiple regression: Factors associated with Agility

A multiple linear regression model was fitted with Agility as the outcome and Acceleration, Sprint.Speed, Balance, Reactions, Stamina, and Strength as predictors. Based on the supplied output, the model R^2 is 0.770, adjusted R^2 is 0.770, and n = 18,979. Thus, the model explains approximately 77.0% of the observed variation in Agility in this dataset.

Predictor	Coefficient	p-value (provided output)	Direction
Acceleration	0.420	Displayed as 0	Positive
Sprint.Speed	0.0927	1.08×10^{-22}	Positive
Balance	0.333	Displayed as 0	Positive
Reactions	0.207	8.96×10^{-226}	Positive
Stamina	0.0673	1.88×10^{-41}	Positive
Strength	-0.0995	1.14×10^{-71}	Negative

All six predictor p-values in the supplied output are below 0.05. R may display extremely small p-values as 0 because of display precision; this should not be interpreted as an exact probability of zero. Holding the other predictors constant, five coefficients are positive and the Strength coefficient is negative. Because player attributes may be correlated and may overlap conceptually, coefficients should be interpreted as conditional associations, not causal effects.

	Relationship	term	estimate	std.error	statistic	p.value	R_squared	Adjusted_R_squared	Model_p_value	Observations
1	Multiple Regression -> Agility	(Intercept)	-1.44355077	0.509062782	-2.835703	4.577335e-03	0.7704242	0.7703516	0	18979
2	Multiple Regression -> Agility	Acceleration	0.42007572	0.010041755	41.832897	0.000000e+00	0.7704242	0.7703516	0	18979
3	Multiple Regression -> Agility	Sprint.Speed	0.09267384	0.009440033	9.817110	1.076282e-22	0.7704242	0.7703516	0	18979
4	Multiple Regression -> Agility	Balance	0.33312483	0.005938639	56.094476	0.000000e+00	0.7704242	0.7703516	0	18979
5	Multiple Regression -> Agility	Reactions	0.20661804	0.006353906	32.518274	8.959241e-226	0.7704242	0.7703516	0	18979
6	Multiple Regression -> Agility	Stamina	0.06734165	0.004981333	13.518800	1.884843e-41	0.7704242	0.7703516	0	18979
7	Multiple Regression -> Agility	Strength	-0.09951927	0.005535567	-17.978153	1.136703e-71	0.7704242	0.7703516	0	18979

Figure 17. Multiple Regression Model Output: Coefficient Table for Agility

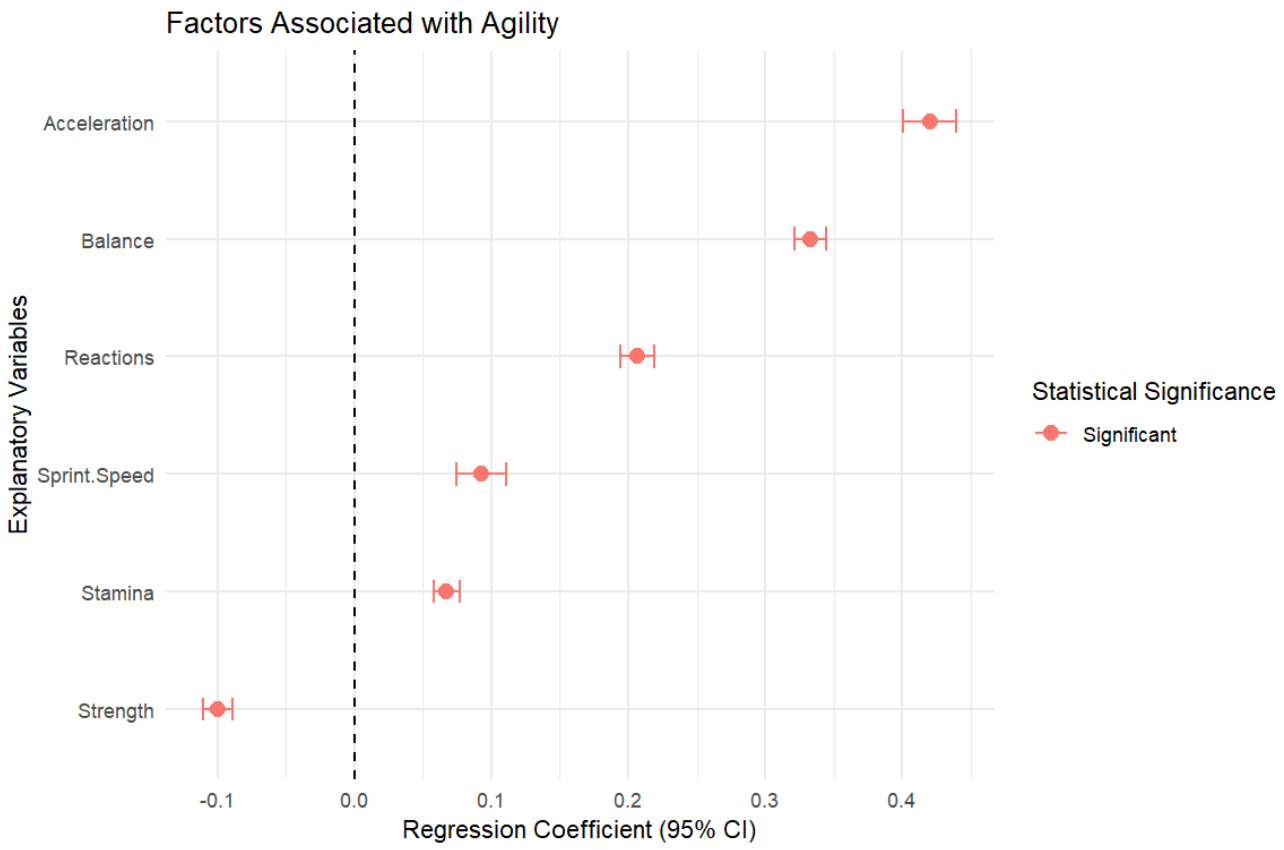


Figure 18. Coefficient Plot for Six Predictors of Agility with 95% Confidence Intervals
A multiple linear regression model was fitted to examine the relationships between six predictors (Acceleration, Sprint Speed, Balance, Reactions, Stamina, and Strength) and player Agility. The model was statistically significant ($p < 0.001$) and explained approximately 77.04% of the variation in Agility ($R^2 = 0.7704$, adjusted $R^2 = 0.7704$). Acceleration ($\beta = 0.420$), Balance ($\beta = 0.333$), Reactions ($\beta = 0.207$), Sprint Speed ($\beta = 0.0927$), and Stamina ($\beta = 0.0673$) had positive coefficients, indicating positive associations with Agility when the other predictors were held constant. In contrast, Strength had a negative coefficient ($\beta = -0.0995$), indicating a negative association with Agility after controlling for the other variables. All six predictors were statistically significant ($p < 0.05$). Overall, the model showed strong explanatory power for Agility in this dataset.

10. Limitations and Recommendations

- This analysis shows relationships between player attributes, but it cannot prove that one attribute causes another to change.
- A low p-value does not always mean a strong relationship. For example, the Height model had a very low R^2.
- Missing values and data errors should be checked before analyzing the data.
- Market value and wages are different, so the difference between them should be interpreted carefully.

- Future analysis could check the model's errors and test its predictions using separate data.
- Some player attributes may be related to each other, which can affect the regression results.

11. Conclusion

This project demonstrates a reproducible workflow for cleaning, exploring, visualizing, and modeling a large player dataset in R. The work standardizes several fields, checks data quality, compares player characteristics by position, screens for players meeting a high-value/low-wage rule, and examines linear relationships among player attributes. The available regression results show a statistically significant but very weak negative association between height and interceptions, while a six-predictor model explains approximately 77% of the variation in agility. The findings are useful for practicing analytical methods and communicating results, but should be interpreted in light of the dataset's structure and the limitations of observational regression.

Appendix A. Suggested Output Checklist

- Confirm Figure 2 shows the populated Missing_Count and Missing_Percentage columns for every variable, not just the column headers.
- Confirm the residual whitespace and newline checks (Figure 3) show zero flagged issues across all character columns after cleaning.
- Verify whether Average_Attacking (Figures 10 and 12) is a single 0–100 rating or a sum of several sub-attributes, and label the axis accordingly so it is not visually compared to Agility or Dribbling on the same 0–100 scale.
- Confirm every figure caption accurately describes what is shown (table vs. chart) and matches its in-text reference.
- Confirm regression diagnostics (linearity, residual normality, homoscedasticity) were reviewed for the Height–Interceptions, Weight–Agility, and multiple regression models.
- Confirm the sample size (n = 18,979) is consistent across all reported models and tables.
- Confirm all reported coefficients, p-values, and R^2 values match the underlying R model output exactly.
- Confirm unit conversions (Height in inches, Weight in kilograms, Value/Wage/Release.Clause in dollars) are applied consistently throughout.
- Proofread the final document for consistent formatting, spelling, and figure numbering.
- Export final summary tables (position_summary, comparison_table, long_term_players, and the high-value/low-wage screening table) to Excel with writexl for supplementary reference.